

Rishi Shah

rrs244@cornell.edu | linkedin.com/in/rishi-shah123 | github.com/rishis123 | rishis123.github.io/rishishah/

EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science (3.7 GPA)

Aug. 2023 – May 2027

- Coursework: Operating Systems, Networks, Data Structures and Algorithms, Databases, Machine Learning, Backend Development, Computer Systems, Functional and Object-Oriented Programming

EXPERIENCE

Microsoft

Aug 2026 – Nov. 2026

SWE Intern, Azure Core

Redmond, WA

- Developed and evaluated an AI-driven incident-intelligence system combining BM25, semantic embeddings, hybrid ranking, and LLM-based relevance adjudication to surface customer-visible CPU performance issues across 180M annual Azure incident records.
- Benchmarked retrieval approaches using precision, recall, and accuracy, and synthesized findings into a research report on reliable incident retrieval.

Apple

May 2026 – Aug. 2026

SWE Intern, Apple Pay

New York City, NY

- Designed a reconciliation and republishing engine (Kotlin/Spring Boot, gRPC) to coordinate on-device deanonymization of payment events, tracking return rates per batch and republishing until analytics thresholds are met.
- Leveraged distributed concurrency controls (Shedlock, mutex) and crash-safe operation ordering across pod failure scenarios to deliver 99.99% reliability at 2,500 TPS sustained, shipping as part of iOS 27.
- Secured distributed infrastructure (DynamoDB, Kafka, Spring Boot on AWS) against gzip bombs, DDoS, malicious payloads, and replay attacks; owned Datadog/Splunk alerting thresholds and observability setup end-to-end.

Capital One

June 2025 – Aug. 2025

SWE Intern

McLean, VA

- Migrated finance ETL from legacy Apache to serverless AWS (Fargate, Batch, Lambda, Step Functions); cut runtime from 8 hrs to 5 min across 500K monthly transactions at 5 companies. First AWS CDK adoption org-wide.
- Built full-stack mentor-matching app using ML vector embeddings (React, TypeScript, Next.js, FastAPI, PostgreSQL, Scikit-learn).

HealthTrends.AI

May 2024 – Aug. 2024

Founding Machine Learning Engineer

New York City, NY

- Built time series models (ARIMA, Prophet, XGBoost) to forecast employee absenteeism; achieved 60% one-month-ahead accuracy, evaluated via MAE, R^2 , and AIC. Presented findings to C-suite to inform product direction.

PROJECTS

Pixel (1st at Google DeepMind × LangChain Multimodal Hacks) | *LangChain, Pinecone, DeepGaze, Gemini*

- Built a multi-agent platform for ad creative optimization: used DeepGaze to model human visual attention and generate scanpath overlays, then applied targeted edits based on predicted attention lift.
- Orchestrated competitor analysis agents over a Pinecone RAG index with LangChain; scored Gemini-generated variants by attention delta against the original creative.

WaveFinder — RAG Music Recommendation System | *ChromaDB, Gemini, TF-IDF, Spotify API*

- Built a music recommendation engine with hybrid retrieval: semantic vector search (ChromaDB) for mood/vibe queries and TF-IDF + popularity-weighted ranking for chart-based queries, routed via a heuristic classifier.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C, C#, JavaScript, TypeScript, OCaml, SQL

Frameworks: Springboot, MERN (JS full-stack), Flask, postgresSQL, Scikit-learn, Pandas

Tools: AWS (DynamoDB, Lambda, Step Functions, Fargate, S3, Batch), Azure, Apache (Kafka, Iceberg), Datadog, LangChain, Pinecone, Splunk, CI/CD, Docker, Git/GitOps, Gradle, Firebase

ON-CAMPUS

Cornell RL Seminar

July 2026 – Present

- Read and discuss reinforcement learning research papers weekly, examining post-training LLMs, GRPO, reward design, systems-level RL infrastructure, and alignment research.

Teaching Assistant – CS 4852: Networks II (Market Design)

Jan. 2024 – Jan. 2026

Resident Advisor

Jan. 2024 – Jan. 2026

ACHIEVEMENTS

National Merit, CAPTRUST, H&R Block Scholars. Google Deepmind x Langchain, Cornell Fintech Hacks Winners. AIME qualifier.